

COLLOQUIO N° 8 : TERMOQUIMICA

Problema N° 1

El gas, al calentarse aumenta su presión

$$m = 10 \text{ g}$$

$$t_1 = 30^\circ$$

$$t_2 = 100^\circ$$

$$W = P \cdot \Delta V \quad \sim \quad V = \text{cte} \Rightarrow \Delta V = 0$$

$$\boxed{W = 0}$$

$$b) \quad Q = C_v \cdot m \cdot \Delta t = \underset{\frac{\text{J}}{\text{g}^\circ\text{C}}}{1,090} \cdot 10 \cdot 10^{-3} \text{ kg} \cdot (100 - 30)^\circ\text{C}$$

$$\boxed{Q = 0,735 \text{ J}}$$

$$c) \quad \Delta E = q + W \Rightarrow \Delta E = q + 0 \Rightarrow \boxed{\Delta E = 0,735 \text{ J}}$$

Problema N° 2

$$\Delta E = q + w$$

Potencia $\rightarrow P = \frac{W}{t} = 200 \frac{J}{s}$ es (-)

Porque entrega W a los alrededores

Calor $2000 \frac{J}{min} \cdot \frac{1 min}{60 seg} = 33,3 \frac{J}{s}$ (-)

Porque "libera calor"

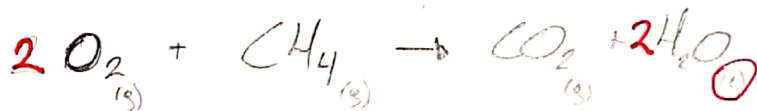
$$\Delta E = -200 \frac{J}{s} - 33,3 \frac{J}{s} \Rightarrow$$

$$\Delta E = -233,3 \frac{J}{s}$$

~ Pierde 233,3 J por segundo !!!

Problema N° 3

Calor de combustión del Metano



$$\Delta H_f^\circ CO_{2(g)} = -393,5 \text{ kJ/mol}$$

$$\Delta H_f^\circ H_2O_{(l)} = -285,8 \text{ kJ/mol}$$

$$\Delta H_f^\circ CH_{4(g)} = -74,81 \text{ kJ/mol}$$

$$\Delta H^\circ_{\text{reacción}} = \sum n \cdot \Delta H_f^\circ \text{ productos} - \sum m \Delta H_f^\circ \text{ reactivos}$$

$$\Delta H^\circ_{\text{reacción}} = \underbrace{(-393,5)}_{CO_2} + \underbrace{(-285,8 \cdot 2)}_{2 \cdot H_2O} - \underbrace{(0)}_{O_2} - \underbrace{(-74,81)}_{CH_4}$$

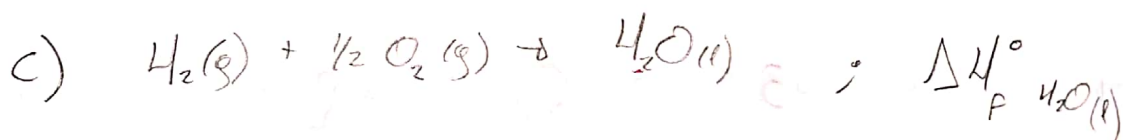
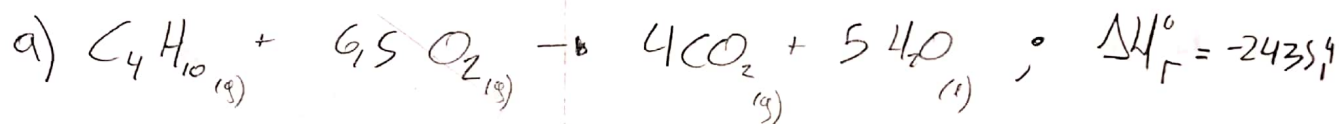
$$\Delta H^\circ_{\text{reacción}} = -890,29 \text{ kJ}$$

(-) $\rightarrow$ es

Exotérmica

Problema N°4

Hallar ΔH_f° del BUTANO C_4H_{10}



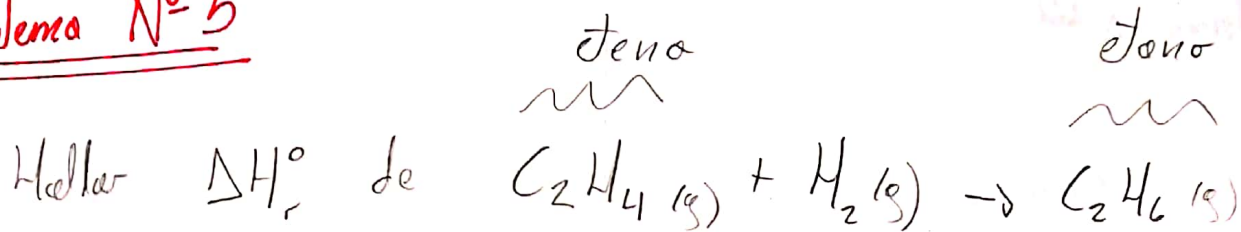
de a) podemos ver que $\Delta H_f^\circ = (4 \Delta H_f^\circ CO_2 + 5 \Delta H_f^\circ H_2O) - \Delta H_f^\circ C_4H_{10}$

$$\Rightarrow \Delta H_f^\circ C_4H_{10(g)} = 4 \cdot \Delta H_f^\circ CO_2 + 5 \Delta H_f^\circ H_2O - \Delta H_f^\circ$$

$$\Delta H_f^\circ C_4H_{10(g)} = -4 \cdot 393,5 - 5 \cdot 285,8 + 2435,4$$

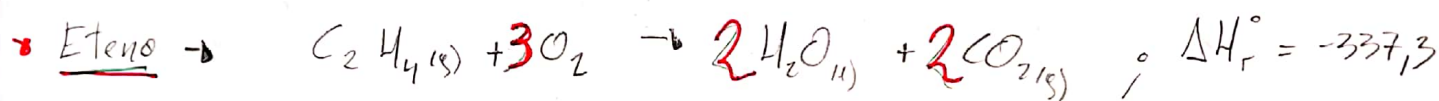
$$\boxed{\Delta H_f^\circ C_4H_{10} = -567,6 \text{ kJ}}$$

Problema N° 5



$\Rightarrow$ Necesito conocer los valores de formación del eteno y el etano.

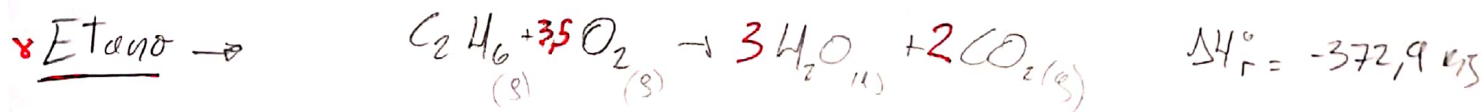
Los hallo de manera indirecta a través de sus valores de combustión!!



$$\Delta H_r^\circ = \underbrace{\left[2\Delta H_f^\circ \text{H}_2\text{O}_{(l)} + 2\Delta H_f^\circ \text{CO}_{2(g)} \right]}_{\text{Productos}} - \underbrace{\left[\Delta H_f^\circ \text{C}_2\text{H}_4(g) + 3\cancel{\Delta H_f^\circ \text{O}_{2(g)}} \right]}_{\text{Reactivos}}$$

$$\Rightarrow \Delta H_f^\circ \text{C}_2\text{H}_4(g) = 2\Delta H_f^\circ \text{H}_2\text{O}_{(l)} + 2\Delta H_f^\circ \text{CO}_{2(g)} - \Delta H_r^\circ$$

$$\Delta H_f^\circ \text{C}_2\text{H}_4(g) = -2 \cdot 285,8 - 2 \cdot 393,5 + 337,3 = \boxed{-1021,3 \text{ kJ}}$$



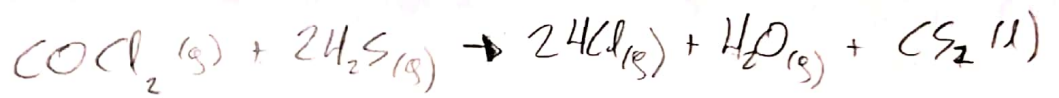
$$\Delta H_f^\circ \text{C}_2\text{H}_6 = 3\Delta H_f^\circ \text{H}_2\text{O} + 2\Delta H_f^\circ \text{CO}_2 - \Delta H_{r\text{Etano}}$$

$$\Delta H_f^\circ \text{C}_2\text{H}_6 = -3 \cdot 285,8 - 2 \cdot 393,5 + 372,9 = \boxed{-1271,5 \text{ kJ}}$$

finalmente $\rightarrow \Delta H_r^\circ = \Delta H_f^\circ \text{C}_2\text{H}_6 - \Delta H_f^\circ \text{C}_2\text{H}_4(g) = -1271,5 + 1021,3$

$$\boxed{\Delta H_r^\circ = -250,2 \text{ kJ}}$$

Problema N° 6



$$\Delta H_r^\circ = \left[2 \Delta H_f^\circ \text{HCl}(g) + \Delta H_f^\circ \text{H}_2\text{O}(g) + \Delta H_f^\circ \text{CS}_2(l) \right] - \left(\Delta H_f^\circ \text{COCl}_2(g) + 2 \Delta H_f^\circ \text{H}_2\text{S}(g) \right)$$

$$\Delta H_r^\circ = (2 \cdot 22,06 - 57,8 + 21,9) - (-53,3 - 4,82)$$

$$\Delta H_r^\circ = -22,8 \text{ kJ}$$